

GNN-Based Network Intrusion Detection

Evaluation Results (v2.1) — XGBoost vs. E-GraphSAGE vs. HeteroGNN vs. EGAT — UNSW-NB15 and CICIDS Test Sets

Summary — All Models Across Datasets

Dataset	Model	Best Val F1	Test Macro-F1	Weighted-F1	MCC	Kappa	Train Time	Status
UNSW	XGBoost	0.5664	0.5910	0.9836	0.8535	0.8534	12.3 min	Complete
	E-GraphSAGE	0.4994	0.5015	0.9806	0.8175	0.8174	6.3 min	Complete
	HeteroGNN	0.5019	0.4976	0.9811	0.8244	0.8242	7.5 min	Complete
	EGAT	0.3997	0.3734	0.9669	0.7153	0.7133	7.6 min	Complete
CICIDS	XGBoost	0.6984	0.6279	0.9862	0.9466	0.9462	73.0 min	Complete
	E-GraphSAGE	0.6918	0.6420	0.9835	0.9314	0.9309	49.0 min	Complete
	HeteroGNN	0.6976	0.5992	0.9813	0.9116	0.9113	67.9 min	Complete
	EGAT	—	—	—	—	—	—	Failed (CUDA OOM)

Accuracy is reported per model inside each dataset section but is omitted here since it is misleading under severe class imbalance; Macro-F1 and per-class F1 are the primary comparison metrics. Four modeling approaches were evaluated: a gradient-boosted tree baseline (XGBoost) and three GNN architectures (E-GraphSAGE, HeteroGNN, EGAT). This revision (v2.1) reflects rerun results for XGBoost, E-GraphSAGE, and HeteroGNN on both datasets; EGAT figures are carried over from the prior run, with its CICIDS run still not completing due to a CUDA out-of-memory error (see CICIDS section).

Version Comparison — v1 -> v2 -> v2.1

v1 covered only E-GraphSAGE and HeteroGNN. v2 added the XGBoost baseline and EGAT, and reran HeteroGNN. v2.1 reran XGBoost, E-GraphSAGE, and HeteroGNN again; EGAT is carried over unchanged from v2 (CICIDS still failing with CUDA OOM). The Chg. Macro-F1 column is the earliest-to-latest available version for each model.

UNSW — UNSW-NB15

Model	Version	Val F1	Macro-F1	Weighted-F1	Train Time	Chg. Macro-F1
XGBoost	v2	0.5647	0.5866	0.9836	18.0 min	
	v2.1	0.5664	0.5910	0.9836	12.3 min	+0.0044
E-GraphSAGE	v1	0.5097	0.5022	0.9815	9.2 min	
	v2	0.5097	0.5022	0.9815	9.2 min	
	v2.1	0.4994	0.5015	0.9806	6.3 min	-0.0007
HeteroGNN	v1	0.4743	0.4882	0.9814	8.7 min	
	v2	0.5013	0.5074	0.9811	9.1 min	
	v2.1	0.5019	0.4976	0.9811	7.5 min	+0.0094
EGAT	v2	0.3997	0.3734	0.9669	7.6 min	
	v2.1	0.3997	0.3734	0.9669	7.6 min (carried over, unchanged)	+0.0000

CICIDS — CIC-IDS2018

Model	Version	Val F1	Macro-F1	Weighted-F1	Train Time	Chg. Macro-F1
XGBoost	v2	0.7000	0.6331	0.9863	129.2 min	
	v2.1	0.6984	0.6279	0.9862	73.0 min	-0.0052
E-GraphSAGE	v1	0.6965	0.6380	0.9835	79.5 min	
	v2	0.6965	0.6380	0.9835	79.5 min	
	v2.1	0.6918	0.6420	0.9835	49.0 min	+0.0040
HeteroGNN	v1	0.7004	0.6203	0.9840	39.3 min	
	v2	0.7041	0.6244	0.9804	76.8 min	
	v2.1	0.6976	0.5992	0.9813	67.9 min	-0.0211
EGAT	v2	—	—	—	Failed (CUDA OOM)	
	v2.1	—	—	—	Failed (CUDA OOM) — unchanged	

UNSW — UNSW-NB15

XGBoost (early stopping at Round 275)

Accuracy	Macro-F1	Weighted-F1	MCC	Cohen's Kappa
0.9846	0.5910	0.9836	0.8535	0.8534
ROC-AUC (macro)	ROC-AUC (micro)	PR-AUC (macro)	Best Val F1	Training Time
0.9939	0.9999	0.6160	0.5664	12.3 min

Per-Class Performance

Class	Prec	Recall	F1	Support	ROC-AUC	PR-AUC
Analysis	0.2400	0.2595	0.2494	185	0.9992	0.2440
Backdoor	0.0526	0.0029	0.0054	700	0.9662	0.0271
Benign	1.0000	1.0000	1.0000	333,440	1.0000	1.0000
DoS	0.7525	0.2508	0.3763	897	0.9916	0.4198
Exploits	0.7259	0.7460	0.7358	6,413	0.9957	0.8185
Fuzzers	0.6606	0.8608	0.7475	5,073	0.9964	0.6885
Generic	0.8413	0.8338	0.8375	2,949	0.9982	0.8549
Reconnaissance	0.8061	0.6038	0.6905	2,562	0.9959	0.7388
Shellcode	0.5144	0.7989	0.6258	358	0.9972	0.7239
Worms	0.6071	0.6800	0.6415	25	0.9988	0.6441

E-GraphSAGE (early stopping at Epoch 57)

Accuracy	Macro-F1	Weighted-F1	MCC	Cohen's Kappa
0.9808	0.5015	0.9806	0.8175	0.8174
ROC-AUC (macro)	ROC-AUC (micro)	PR-AUC (macro)	Best Val F1	Training Time
0.9828	0.9995	0.4866	0.4994	6.3 min

Per-Class Performance

Class	Prec	Recall	F1	Support	ROC-AUC	PR-AUC
Analysis	0.3432	0.8811	0.4939	185	0.9922	0.4717
Backdoor	0.0039	0.0043	0.0041	700	0.8823	0.0181
Benign	0.9999	0.9996	0.9998	333,440	1.0000	1.0000
DoS	0.3043	0.0858	0.1339	897	0.9873	0.1907

Class	Prec	Recall	F1	Support	ROC-AUC	PR-AUC
Exploits	0.7286	0.6510	0.6876	6,413	0.9943	0.7652
Fuzzers	0.6438	0.8328	0.7262	5,073	0.9951	0.6430
Generic	0.8603	0.7497	0.8012	2,949	0.9950	0.8066
Reconnaissance	0.7091	0.5593	0.6254	2,562	0.9932	0.6729
Shellcode	0.2457	0.5922	0.3473	358	0.9910	0.2046
Worms	0.2500	0.1600	0.1951	25	0.9974	0.0934

GNN Structural Metrics

Avg. Nodes/Graph	Avg. Edges/Graph	Neighbor Coverage	Silhouette	Davies-Bouldin
1,918	1,671	71.95%	0.2430	N/A

HeteroGNN (early stopping at Epoch 58)

Accuracy	Macro-F1	Weighted-F1	MCC	Cohen's Kappa
0.9815	0.4976	0.9811	0.8244	0.8242
ROC-AUC (macro)	ROC-AUC (micro)	PR-AUC (macro)	Best Val F1	Training Time
0.9051	0.9978	0.5167	0.5019	7.5 min

Per-Class Performance

Class	Prec	Recall	F1	Support	ROC-AUC	PR-AUC
Analysis	0.2873	0.9784	0.4442	185	0.9891	0.5016
Backdoor	0.0000	0.0000	0.0000	700	0.3708	0.0055
Benign	1.0000	0.9999	1.0000	333,440	1.0000	1.0000
DoS	0.2442	0.1048	0.1466	897	0.9818	0.1425
Exploits	0.7805	0.6454	0.7066	6,413	0.9919	0.7717
Fuzzers	0.6467	0.8384	0.7302	5,073	0.9763	0.6606
Generic	0.7562	0.7982	0.7766	2,949	0.9843	0.7933
Reconnaissance	0.7602	0.5480	0.6369	2,562	0.9930	0.6916
Shellcode	0.2321	0.6788	0.3459	358	0.8386	0.4723
Worms	0.1786	0.2000	0.1887	25	0.9252	0.1277

GNN Structural Metrics

Avg. Nodes/Graph	Avg. Edges/Graph	Neighbor Coverage	Silhouette	Davies-Bouldin
1,703	6,684	100.00%	0.7727	N/A

EGAT (early stopping at Epoch 57)

Accuracy	Macro-F1	Weighted-F1	MCC	Cohen's Kappa
0.9697	0.3734	0.9669	0.7153	0.7133
ROC-AUC (macro)	ROC-AUC (micro)	PR-AUC (macro)	Best Val F1	Training Time
0.9915	0.9996	0.4856	0.3997	7.6 min

Per-Class Performance

Class	Prec	Recall	F1	Support	ROC-AUC	PR-AUC
Analysis	0.4606	0.6324	0.5330	185	0.9941	0.6370
Backdoor	0.0000	0.0000	0.0000	700	0.9731	0.0376
Benign	1.0000	0.9994	0.9997	333,440	0.9999	1.0000
DoS	0.0000	0.0000	0.0000	897	0.9868	0.1257
Exploits	0.6515	0.7146	0.6816	6,413	0.9934	0.7408
Fuzzers	0.5810	0.0240	0.0462	5,073	0.9926	0.5453
Generic	0.2437	0.7918	0.3727	2,949	0.9908	0.5102
Reconnaissance	0.9009	0.4863	0.6317	2,562	0.9943	0.6931
Shellcode	0.3341	0.7877	0.4692	358	0.9941	0.5508
Worms	0.0000	0.0000	0.0000	25	0.9956	0.0152

GNN Structural Metrics

Avg. Nodes/Graph	Avg. Edges/Graph	Neighbor Coverage	Silhouette	Davies-Bouldin
1,918	1,671	71.95%	0.5195	N/A

CICIDS — CIC-IDS2018

XGBoost (early stopping at Round 192)

Accuracy	Macro-F1	Weighted-F1	MCC	Cohen's Kappa
0.9894	0.6279	0.9862	0.9466	0.9462
ROC-AUC (macro)	ROC-AUC (micro)	PR-AUC (macro)	Best Val F1	Training Time
0.9867	0.9996	0.6404	0.6984	73.0 min

Per-Class Performance

Class	Prec	Recall	F1	Support	ROC-AUC	PR-AUC
Benign	0.9950	0.9996	0.9973	2,608,914	0.9914	0.9987
Bot	0.9998	0.9959	0.9978	16,719	0.9999	0.9966
Brute_Force_-Web	0.0526	0.0041	0.0076	244	0.9977	0.0927
Brute_Force_-XSS	0.0000	0.0000	0.0000	69	0.9998	0.0684
DDoS_attack-HOIC	1.0000	0.9998	0.9999	154,848	1.0000	1.0000
DDoS_attack-LOIC-UDP	0.2078	0.6385	0.3135	260	0.9724	0.1725
DDoS_attacks-LOIC-HTTP	0.9970	0.9818	0.9893	43,289	0.9856	0.9855
DoS_attacks-GoldenEye	0.9804	0.9998	0.9900	4,598	1.0000	0.9999
DoS_attacks-Hulk	0.9980	0.9987	0.9984	15,012	0.9998	0.9994
DoS_attacks-SlowHTTPTest	0.0000	0.0000	0.0000	15,833	0.9950	0.3531
DoS_attacks-Slowloris	0.9823	0.8206	0.8942	2,703	0.9972	0.9185
FTP-BruteForce	0.6469	1.0000	0.7856	29,004	0.9973	0.6469
Infiltration	0.8550	0.3016	0.4459	17,372	0.8644	0.3236
SQL_Injection	0.0000	0.0000	0.0000	66	0.9998	0.0500
SSH-Bruteforce	1.0000	0.9993	0.9996	14,137	1.0000	1.0000

E-GraphSAGE (early stopping at Epoch 56)

Accuracy	Macro-F1	Weighted-F1	MCC	Cohen's Kappa
0.9865	0.6420	0.9835	0.9314	0.9309
ROC-AUC (macro)	ROC-AUC (micro)	PR-AUC (macro)	Best Val F1	Training Time
0.9871	0.9994	0.6562	0.6918	49.0 min

Per-Class Performance

Class	Prec	Recall	F1	Support	ROC-AUC	PR-AUC
Benign	0.9930	0.9983	0.9956	2,608,914	0.9858	0.9976
Bot	0.9999	0.9977	0.9988	16,719	0.9990	0.9977
Brute_Force_-Web	0.1952	0.4713	0.2761	244	0.9990	0.1442
Brute_Force_-XSS	0.0665	0.9565	0.1244	69	0.9999	0.0997
DDOS_attack-HOIC	0.9999	1.0000	1.0000	154,848	1.0000	1.0000
DDOS_attack-LOIC-UDP	0.2719	0.9423	0.4220	260	0.9894	0.1768
DDoS_attacks-LOIC-HTTP	1.0000	0.9487	0.9737	43,289	0.9831	0.9829
DoS_attacks-GoldenEye	1.0000	0.7747	0.8730	4,598	0.9998	0.9061
DoS_attacks-Hulk	0.9999	0.8372	0.9114	15,012	1.0000	0.9985
DoS_attacks-SlowHTTPTest	0.0000	0.0000	0.0000	15,833	0.9943	0.3251
DoS_attacks-Slowloris	1.0000	0.8302	0.9072	2,703	1.0000	1.0000
FTP-BruteForce	0.6469	1.0000	0.7856	29,004	0.9969	0.6139
Infiltration	0.5767	0.2644	0.3626	17,372	0.8591	0.2839
SQL_Injection	0.0000	0.0000	0.0000	66	0.9998	0.3158
SSH-Bruteforce	1.0000	1.0000	1.0000	14,137	1.0000	1.0000

GNN Structural Metrics

Avg. Nodes/Graph	Avg. Edges/Graph	Neighbor Coverage	Silhouette	Davies-Bouldin
3,389	3,051	93.69%	0.1327	N/A

HeteroGNN (early stopping at Epoch 65)

Accuracy	Macro-F1	Weighted-F1	MCC	Cohen's Kappa
0.9820	0.5992	0.9813	0.9116	0.9113
ROC-AUC (macro)	ROC-AUC (micro)	PR-AUC (macro)	Best Val F1	Training Time
0.9918	0.9996	0.6721	0.6976	67.9 min

Per-Class Performance

Class	Prec	Recall	F1	Support	ROC-AUC	PR-AUC
Benign	0.9955	0.9921	0.9938	2,608,914	0.9971	0.9996
Bot	0.9996	0.9743	0.9868	16,719	0.9996	0.9978
Brute_Force_-Web	0.2353	0.0164	0.0307	244	0.9989	0.2734
Brute_Force_-XSS	0.0741	0.0290	0.0417	69	0.9991	0.1251

Class	Prec	Recall	F1	Support	ROC-AUC	PR-AUC
DDoS_attack-HOIC	1.0000	1.0000	1.0000	154,848	1.0000	1.0000
DDoS_attack-LOIC-UDP	0.2554	0.9038	0.3983	260	0.9907	0.2522
DDoS_attacks-LOIC-HTTP	1.0000	0.9487	0.9737	43,289	0.9841	0.9827
DoS_attacks-GoldenEye	0.4862	0.7749	0.5975	4,598	0.9997	0.8947
DoS_attacks-Hulk	1.0000	0.8372	0.9114	15,012	1.0000	0.9997
DoS_attacks-SlowHTTPTest	0.0000	0.0000	0.0000	15,833	0.9953	0.4184
DoS_attacks-Slowloris	1.0000	0.8391	0.9125	2,703	1.0000	0.9666
FTP-BruteForce	0.6469	1.0000	0.7856	29,004	0.9974	0.6782
Infiltration	0.2853	0.4758	0.3567	17,372	0.9563	0.4373
SQL_Injection	0.0000	0.0000	0.0000	66	0.9586	0.0561
SSH-Bruteforce	0.9998	1.0000	0.9999	14,137	1.0000	1.0000

GNN Structural Metrics

Avg. Nodes/Graph	Avg. Edges/Graph	Neighbor Coverage	Silhouette	Davies-Bouldin
3,566	12,205	100.00%	0.3558	N/A

EGAT — Training Failed

Training on CICIDS was aborted during the edge-attention update step (GATv2Conv) with a CUDA out-of-memory error — the process had allocated roughly 14.2 GiB against a 14.56 GiB GPU and could not reserve an additional ~700 MiB batch of edge attention weights. No test-set metrics are available for this configuration; UNSW-NB15 training completed without issue on the same hardware.